

Grade 6 Accelerated
Unit 14
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

Write and solve an algebraic equation for each of the following.

1. At a restaurant, 6 friends decided to split the bill evenly. If each person paid \$14, what was the total bill?

$$\frac{t}{6} = \$14 \quad t = \$84 \quad \text{The total bill was \$84.}$$

2. A Great White Shark has 300 teeth. If the teeth are evenly arranged into 3 rows of teeth, how many teeth are in each row?

$$\frac{300}{t} = 3 \quad t = 100 \quad \text{Each row has 100 teeth.}$$

3. A school has a budget of \$400,000 for support staff members. If each support staff member makes \$40,000 each, how many support staff members can the school employ?

$$\frac{40,000s}{40,000} = \frac{400,000}{40,000} \quad \text{The school can employ 10 support staff members.}$$
$$s = 10$$

4. During a math adoption, there were 4 textbook company finalists invited to present their materials to the school district. For spacing purposes, each company could only bring a maximum of 3 people each. What is the maximum amount of people that could attend from the textbook companies?

$$4 \cdot 3 = t \text{ or } \frac{t}{4} = 3 \quad 12 \text{ people max can attend.}$$
$$t = 12$$

5. A princess dress takes 6 yards of fabric to create. If an order is placed for 9 dresses, how many yards of fabric will be needed?

$$\frac{d}{9} = 6 \quad d = 54 \quad 54 \text{ yards of fabric will be needed.}$$

6. A certain apple tree can produce 210 apples at a given time. If the apples are placed into bins that can hold 21 apples each, how many crates will be needed for the entire tree?

$$\frac{210}{c} = 21 \quad c = 10 \quad 10 \text{ crates will be needed.}$$

7. A recipe for Red Velvet cake calls for 1.5 sticks of butter for each cake. If someone wants to make 4 cakes, how many sticks of butter will be needed?

$$\frac{b}{4} = 1.5 \quad b = 6 \quad 6 \text{ sticks of butter will be needed.}$$

8. A new home builder offers 6 different floorplans, each that come with 5 color pallet options per floorplan. What is the number of different options that a buyer has with this builder?

$$\frac{f}{5} = 6 \quad f = 30 \quad \text{Buyers have 30 different options.}$$

9. A garment moving box can hold 8 suit jackets. If 4 garment boxes can fit in the back of a vehicle, what is the maximum number of suit jackets that can be moved in one trip?

$$\frac{g}{4} = 8 \quad g = 32 \quad 32 \text{ suit jackets can be moved.}$$

10. Thelma is participating in a Walk-a-Thon where she will get \$0.25 per lap that she walks. How many laps would she need to walk to reach a goal of \$50?

$$0.25w = 50 \quad \text{Thelma needs to walk 200 laps.}$$
$$w = 200$$